

state machines

CSCI
270

rules

of word ladder

- ▶ change **one** letter at a time
- ▶ each intermediate word must be valid english

p r a w n

your score is the number
of steps you take

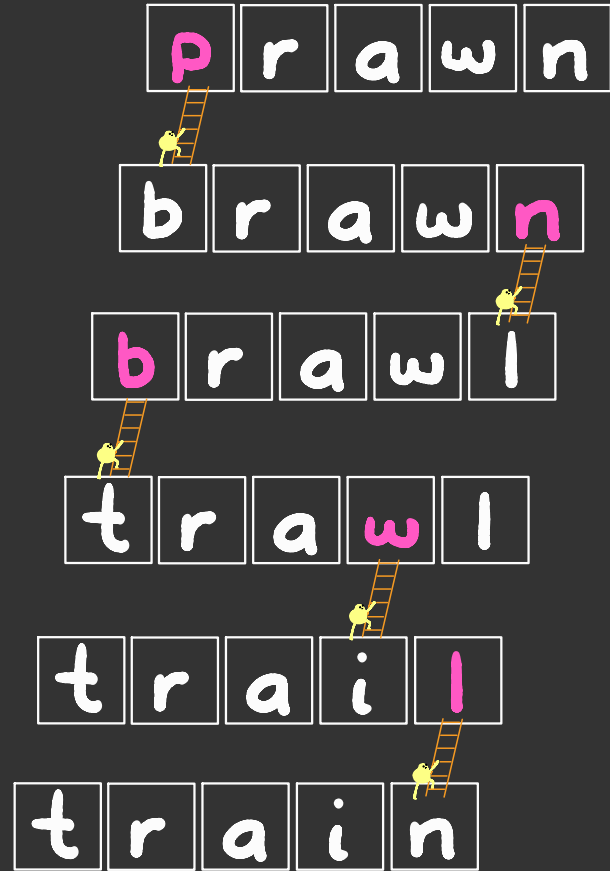


t r a i n

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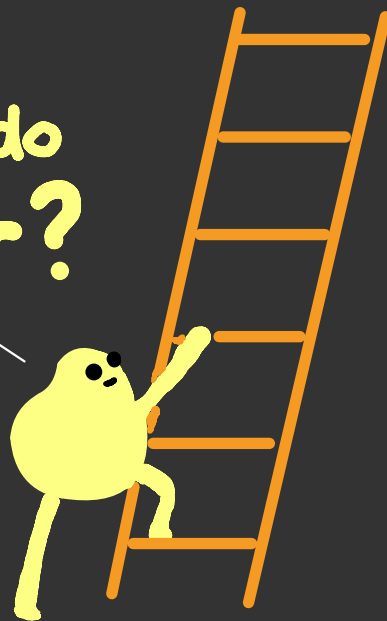
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p r a w n

can you do
better?

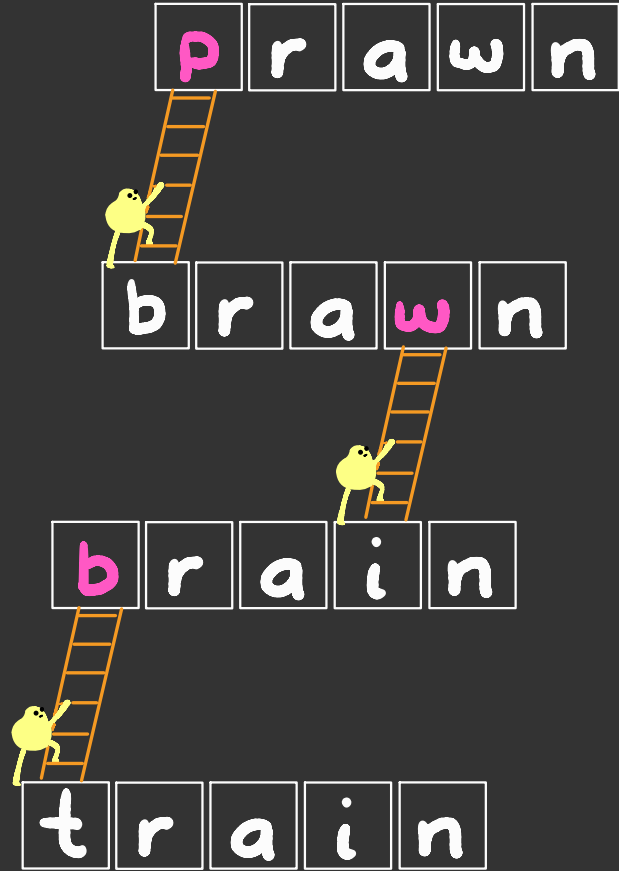


t r a i n

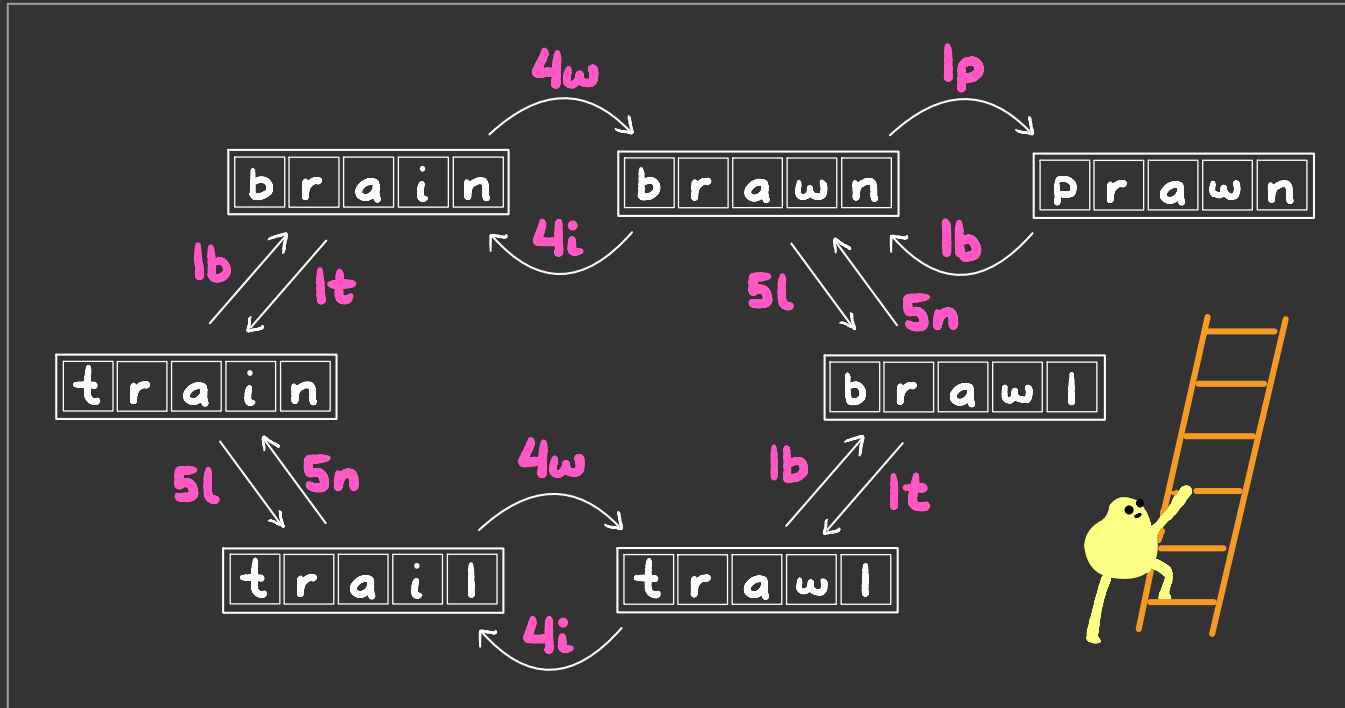
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we can describe word ladder using
a graph called a **state machine**



a state machine is a tuple

$$(Q, \Sigma, \Delta, q_0, F)$$

where :

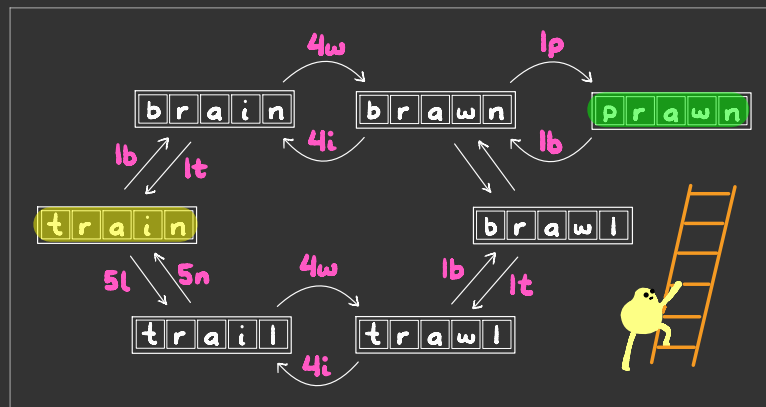
Q is a set of symbols called **states**

Σ is a set of symbols called **actions**

$\Delta \subseteq Q \times \Sigma \times Q$ is a set of triples called **transitions**

$q_0 \in Q$ is the **initial state**

$F \subseteq Q$ is a set of **final states**



e.g.

$Q = \{\text{all 5-letter words}\}$

$\Sigma = \{1a, 1b, 1c, \dots, 5y, 5z\}$

$\Delta = \{(\text{train}, 1b, \text{brain}), \dots\}$

$q_0 = \text{train}$

$F = \{\text{prawn}\}$

a **weighted** state machine is a tuple

$$(Q, \Sigma, \Delta, q_0, F, \omega)$$

where :

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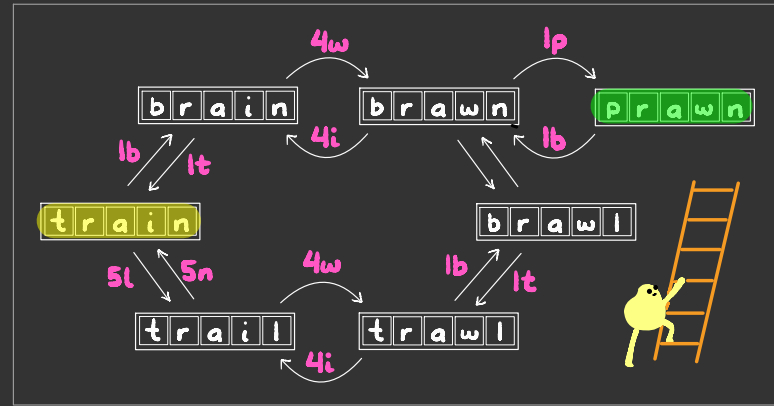
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ω maps each transition to a real-valued **weight**



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$\omega =$

what is the weight function
for word ladder?

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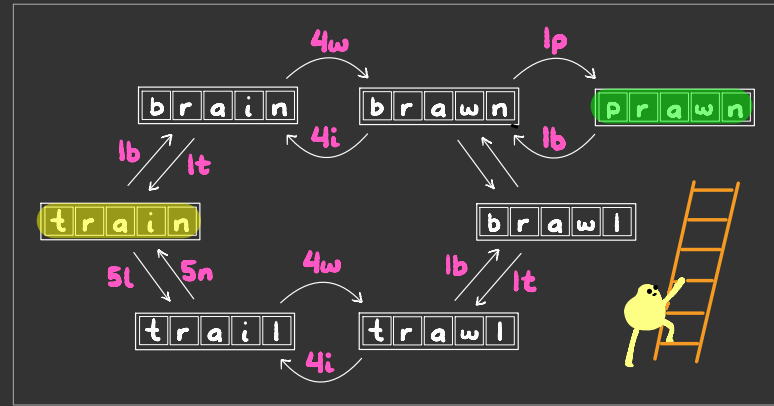
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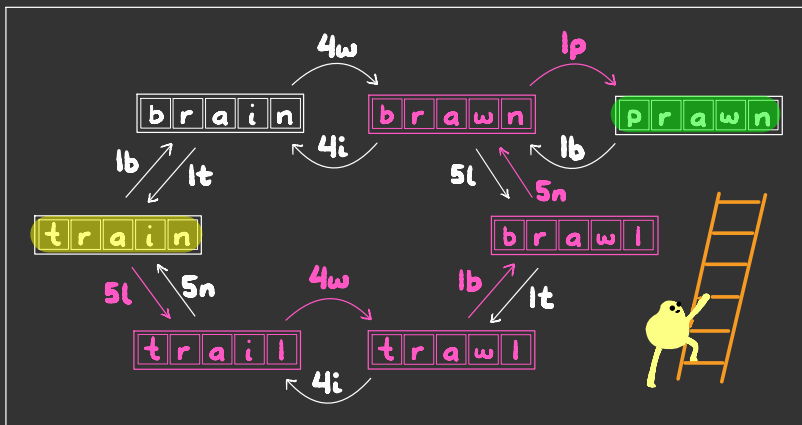
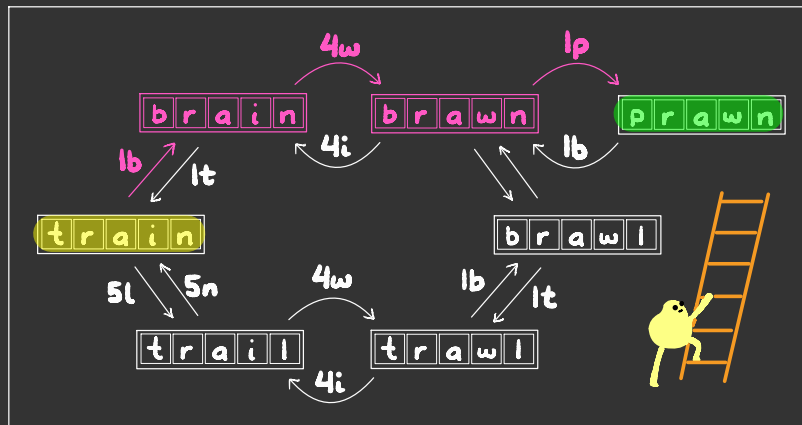
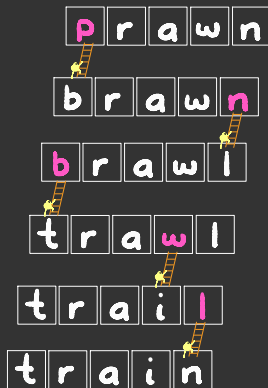
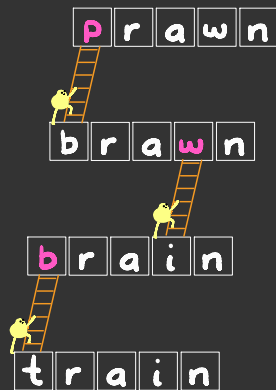
$q_0 = \text{train}$

$F = \{\text{prawn}\}$

$\omega = \{\delta \mapsto 1 \mid \delta \in \Delta\}$

a solution to
word ladder is a
path

in the state machine
from **train** to **prawn**



a search path is a sequence

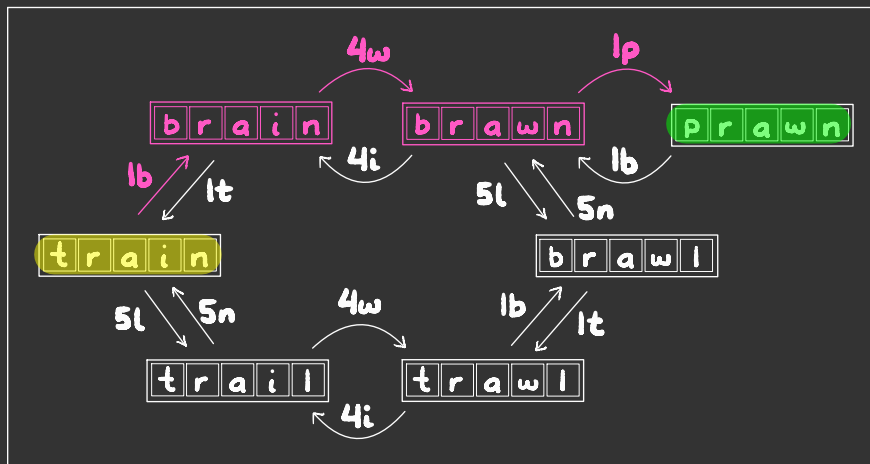
$$[\delta_0, \dots, \delta_k]$$

of transitions such that

there exist states $q_1, \dots, q_{k+1} \in Q$

and actions $\sigma_0, \dots, \sigma_k \in \Sigma$

such that $\delta_i = (q_i, \sigma_i, q_{i+1}) \forall i \in \{0, \dots, k\}$

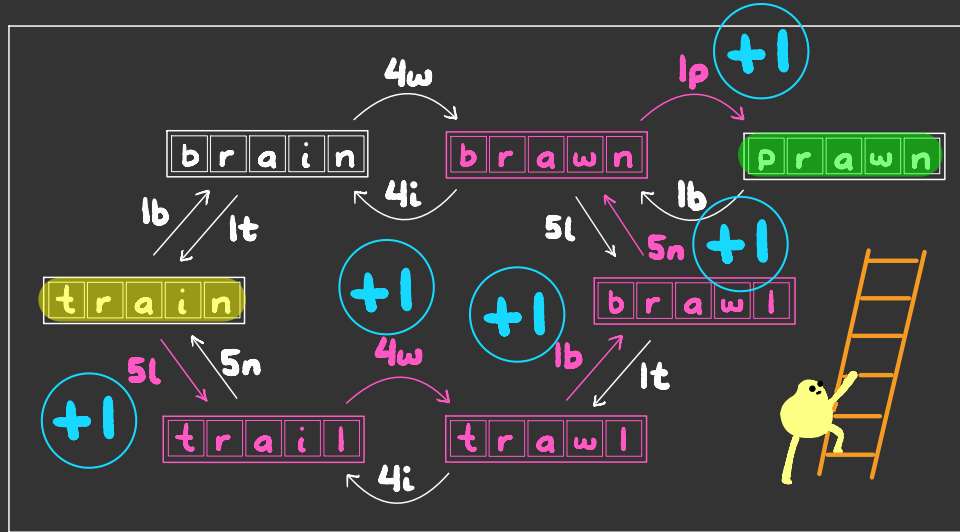


$$\left[\left(\boxed{\text{train}}, \text{lb}, \boxed{\text{brain}} \right), \left(\boxed{\text{brain}}, \text{4w}, \boxed{\text{brawn}} \right), \left(\boxed{\text{brawn}}, \text{lp}, \boxed{\text{prawn}} \right) \right]$$

these must
match

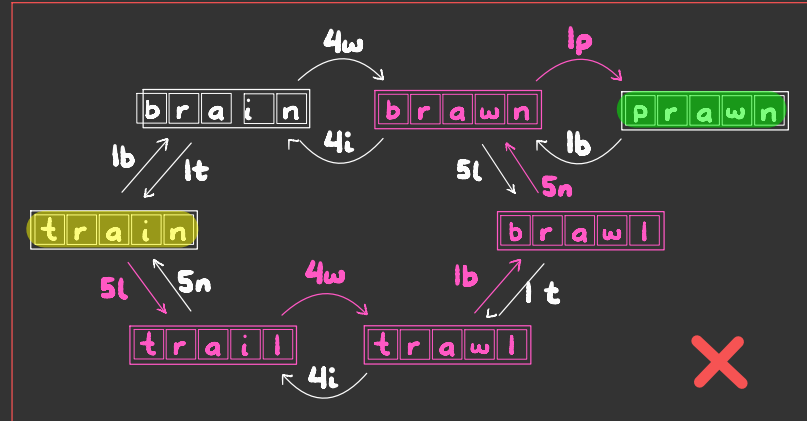
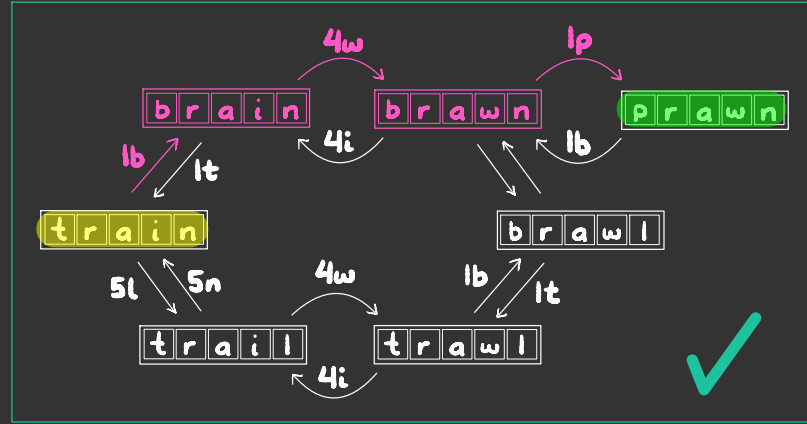
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in a weighted state machine, the **cost** of a search path is the sum of the weights of the transitions

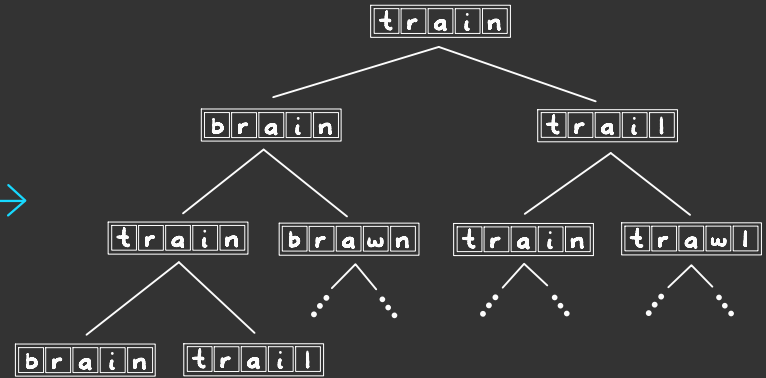
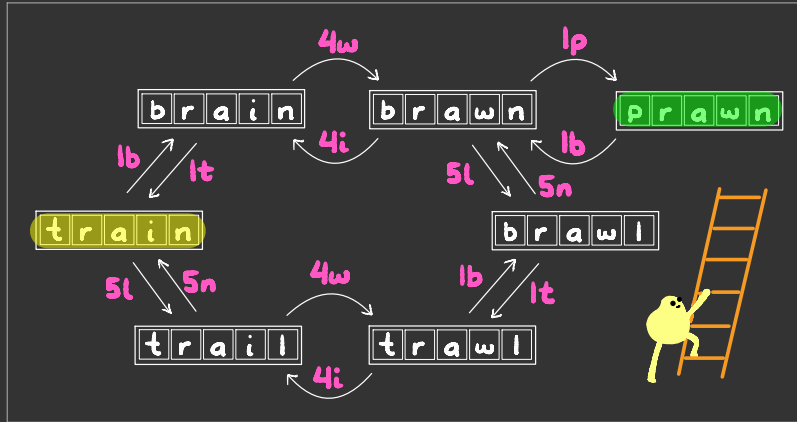


so this path has cost **5**

the goal of search
is to find a path
of **minimal cost**
whose last transition
leads to a **final state**

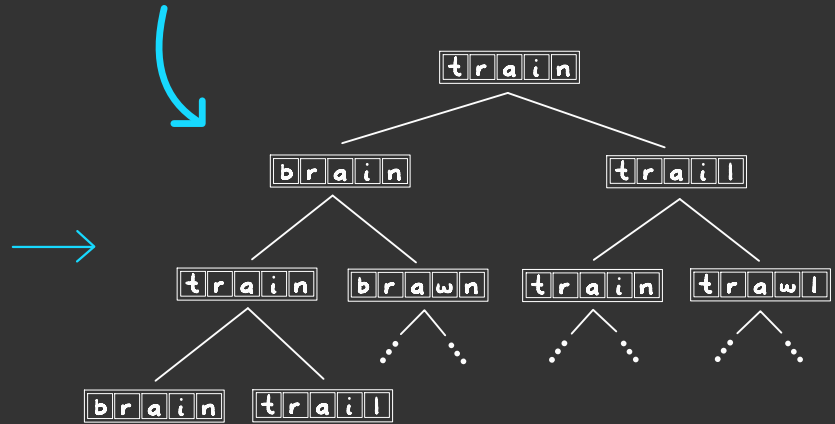
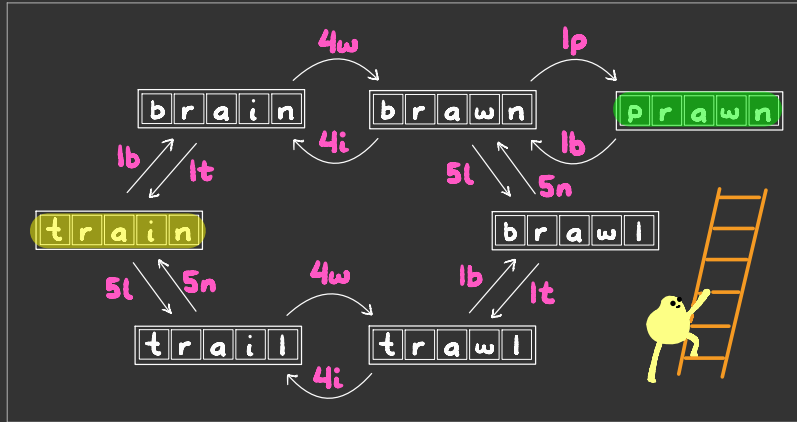


a state machine can be unraveled
into a **search tree**



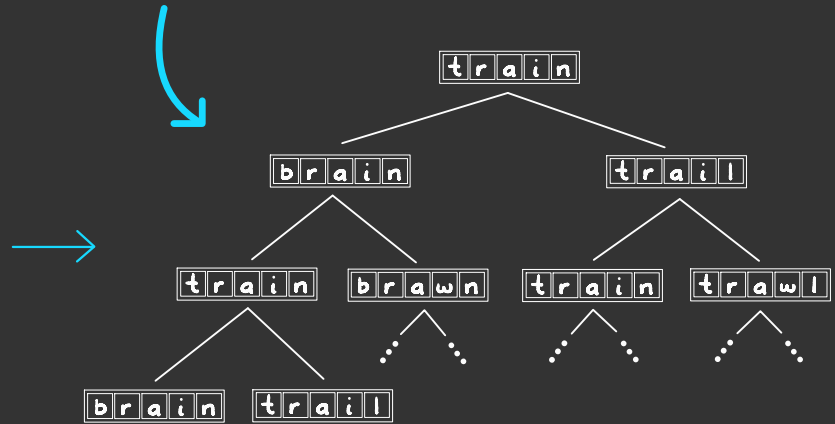
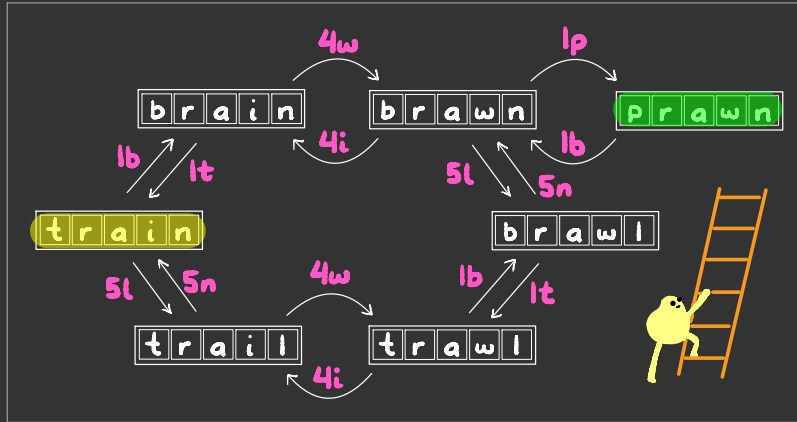
the search tree is a compact
representation of all search paths

what is the **depth**
of this search tree?



???

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of this search tree?



infinite. if a state machine has reachable
cycles, its search tree has infinite depth